

Victor A. Pinto Abarzúa

Assistant Professor, Department of Physics · Universidad de Santiago de Chile
(USACH)
Associate Chair for Teaching (since August 2025)

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Research interests

I study how the solar wind controls the dynamics of Earth's radiation belts: when and why relativistic electrons are accelerated, transported or lost. I combine satellite data (Van Allen Probes, THEMIS, GOES) and ground magnetometers with machine learning models to forecast geomagnetic perturbations and geomagnetically induced currents (GIC).

Education

2019	Ph.D. in Atmospheric and Oceanic Sciences (space physics) , University of California, Los Angeles (UCLA). Co-advisors: Larry Lyons and Jacob Bortnik
2014	M.S. in Atmospheric and Oceanic Sciences , UCLA
2011	M.Sc. in Physics (highest distinction) , Universidad de Chile. Advisors: Alejandro Valdivia and Marina Stepanova
2006	B.Sc. in Physics , Universidad de Chile

Positions

2022–	Assistant Professor , Department of Physics, Universidad de Santiago de Chile (USACH). Associate Chair for Teaching since August 2025
2020–2022	Postdoctoral Research Associate , Space Science Center, University of New Hampshire
2015–2019	Teaching Assistant, Associate and Fellow; Graduate Student Researcher (2017–2019) , Department of Atmospheric and Oceanic Sciences, UCLA
2014, 2016	Visiting researcher , NASA Goddard Space Flight Center
2011	Chilean Antarctic Scientific Expedition 47 , INACH. Replaced and calibrated magnetometers at Escudero and O'Higgins bases
2010–2011	Instructor , Universidad Católica del Maule; Universidad de Valparaíso

Research funding

2025–2028	FONDECYT Initiation grant (ANID) — Principal investigator. ULF waves and trapped particles in the radiation belts
2025–2027	FONDEF IDEA I+D (ANID) — Principal investigator. Pilot space weather platform for Chile. Principal investigator for USACH; led by Universidad Adventista de Chile
2023–	FONDEQUIP (ANID) — Director. High-performance computing (HPC) platform, Department of Physics
2025–2029	FONDECYT Regular (ANID) — Co-investigator. Turbulence and pressure balance in the magnetosphere. PI: M. Stepanova
2024–2028	FONDECYT Regular (ANID) — Co-investigator. Physics of expanding plasmas (solar and stellar winds). PI: P. S. Moya
2025–2029	CNPq (Brazil) (CNPq) — Co-investigator. GIC in Brazil and precipitation in the South Atlantic Anomaly. PI: J. P. Marchezi (INPE)

2023-2027	NASA Living With a Star (NASA) — International collaborator. Foreshock transients and magnetospheric perturbations. PI: J. Liu (UCLA)
2022-2025	Academic Insertion grant (ANID) — Principal investigator
2024-2025	DICYT Regular (USACH) — Principal investigator
2024	DICYT Juventud Investigadora (USACH) — Principal investigator

Peer-reviewed journal articles

1. Coello-Guzmán, M., **Pinto, V.A.**, Navarro, R.E., Moya, P.S. (2026). The effect of expansion and instabilities in the thermodynamic regulation of the young solar wind plasma. *Astronomy & Astrophysics* (in press).
2. Marchezi, J.P., Keesee, A., Coughlan, M., Mukundan, R., **Pinto, V.**, Da Silva, L.A., Alves, L.R., Connor, H., Hampton, D. (2026). On the effects of the solar wind structure in the global distribution of dBH/dt spikes during geomagnetic storms. *JGR: Space Physics*. doi:10.1029/2025JA034994
3. Caro, G., Carrasco, E., Diaz Peña, J., Urra, B., Rojo, E., Castro, L., Bravo, M., **Pinto, V.**, Stepanova, M., Godoy, A., Ovalle, E., Denardini, C.M., Su Chen, S. (2026). Comparative performance analysis of five magnetometers for geomagnetic monitoring. *Sensors*, 26, 4700. doi:10.3390/s26154700
4. Inostroza, A.M., Díaz Peña, J., Coello-Guzmán, M., Stepanova, M., **Pinto, V.A.**, Navarro, R.E., Espinoza, J.M., Kolomiytseva, E.A., Antonova, E.E. (2026). The role of auroral phenomena in the dynamics of the outer radiation belt during the 1 June 2013 geomagnetic storm. *JGR: Space Physics*, 131, e2026JA035123. doi:10.1029/2026JA035123
5. Stepanova, M.V., **Pinto, V.A.**, Espinoza, C.M., Ovchinnikov, I., Antonova, E.E. (2026). Role of the solar wind dynamic pressure in the development of magnetospheric turbulence. *Physica Scripta*, 101(9). doi:10.1088/1402-4896/ae48b2
6. **Pinto, V.A.**, Espitia, Y., Zenteno-Quinteros, B., Stepanova, M.V., Moya, P.S. (2025). Radial evolution of ultrarelativistic electrons during enhancement events. *Physica Scripta*, 100(11). doi:10.1088/1402-4896/ae1332
7. Mukundan, R., Keesee, A.M., Marchezi, J.P., **Pinto, V.A.**, Coughlan, M., Hampton, D. (2025). Localized geomagnetic disturbances: a statistical analysis of spatial scale. *Frontiers in Astronomy and Space Sciences*, 12, 1610276. doi:10.3389/fspas.2025.1610276
8. Coughlan, M., Keesee, A., **Pinto, V.**, Mukundan, R., Marchezi, J.P., et al (2025). Using machine learning explainability techniques to examine drivers of ground magnetic field localization. *Space Weather*, 23, e2025SW004391. doi:10.1029/2025SW004391
9. Lara, C., **Pinto, V.A.**, Silva, J., Zenteno-Quinteros, B., Moya, P.S. (2025). On the Relationship Between ULF Wave Power and Changes of Relativistic Electron Fluxes in the Outer Radiation Belt. *Universe*, 11(5), 151. doi:10.3390/universe11050151
10. Gallardo-Lacourt, B., Nishimura, Y., ..., **Pinto, V.A.**, et al (2024). Unexpected STEVE observations at high latitude during quiet geomagnetic conditions. *Geophysical Research Letters*, 51, e2024GL110568. doi:10.1029/2024GL110568
11. Stepanova, M., **Pinto, V.**, Antonova, E. (2024). Regarding the relativistic electron dynamics in the outer radiation belt: a historical view. *Reviews of Modern Plasma Physics*, 8(1), 25. doi:10.1007/s41614-024-00165-4
12. Coughlan, M., Keesee, A.M., **Pinto, V.A.**, Mukundan, R., Marchezi, J.P., et al (2023). Probabilistic Forecasting of Ground Magnetic Perturbation Spikes at Mid-Latitude Stations. *Space Weather*, 21, e2023SW003446. doi:10.1029/2023SW003446
13. Moya, P.S., Zenteno-Quinteros, B., Gallo-Mendez, I., **Pinto, V.A.** (2022). The role of O⁺ and He⁺ in the propagation of kinetic Alfvén waves in the Earth's inner magnetosphere. *The Astrophysical Journal*, 933, 32. doi:10.3847/1538-4357/ac713b

14. **Pinto, V.A.**, Keesee, A.M., Coughlan, M., Mukundan, R., Johnson, J.W., et al (2022). Revisiting the Ground Magnetic Field Perturbations Challenge: A Machine Learning Perspective. *Frontiers in Astronomy and Space Sciences*, 9, 869740. doi:10.3389/fspas.2022.869740
15. Blandin, M., Connor, H.K., ..., **Pinto, V.A.**, et al (2022). Multi-Variate LSTM Prediction of Alaska Magnetometer Chain Utilizing a Coupled Model Approach. *Frontiers in Astronomy and Space Sciences*, 9, 846291. doi:10.3389/fspas.2022.846291
16. Reyes, P.I., **Pinto, V.A.**, Moya, P.S. (2021). Geomagnetic storm occurrence and their relation with solar cycle phases. *Space Weather*, 19, e2021SW002766. doi:10.1029/2021SW002766
17. Stepanova, M., **Pinto, V.A.**, Antonova, E.E. (2021). Adiabatic and non-adiabatic evolution of relativistic electrons in the heart of the outer radiation belt during the 1 June 2013 geomagnetic storm. *Journal of Atmospheric and Solar-Terrestrial Physics*, 212. doi:10.1016/j.jastp.2020.105479
18. Keesee, A.M., **Pinto, V.A.**, et al (2020). Comparison of Deep Learning Techniques to Model Connections Between Solar Wind and Ground Magnetic Perturbations. *Frontiers in Astronomy and Space Sciences*, 7, 550874. doi:10.3389/fspas.2020.550874
19. **Pinto, V.A.**, Bortnik, J., Moya, P.S., Lyons, L.R., Sibeck, D.G., et al (2020). Radial response of outer radiation belt relativistic electrons during enhancement events at geostationary orbit. *JGR: Space Physics*, 125, e2019JA027660. doi:10.1029/2019JA027660
20. **Pinto, V.A.**, Zhang, X.-J., Mourenas, D., Bortnik, J., Artemyev, A.V., et al (2020). On the confinement of ultrarelativistic electron remnant belts to low L-shells. *JGR: Space Physics*, 125, e2019JA027469. doi:10.1029/2019JA027469
21. **Pinto, V.A.**, Mourenas, D., Bortnik, J., Zhang, X.-J., Artemyev, A.V., et al (2019). Decay of ultrarelativistic remnant belt electrons through scattering by plasmaspheric hiss. *JGR: Space Physics*, 124. doi:10.1029/2019JA026509
22. Gabrielse, C., **Pinto, V.A.**, Nishimura, Y., Lyons, L., et al (2019). Storm time mesoscale plasma flows in the nightside high-latitude ionosphere. *Geophysical Research Letters*, 46, 4079-4088. doi:10.1029/2018GL081539
23. Stepanova, M., Antonova, E.E., Moya, P.S., **Pinto, V.A.**, Valdivia, J.A. (2019). Multisatellite analysis of plasma pressure in the inner magnetosphere during the 1 June 2013 geomagnetic storm. *JGR: Space Physics*, 124, 1187-1202. doi:10.1029/2018JA025965
24. **Pinto, V.A.**, Bortnik, J., Moya, P.S., Lyons, L.R., Sibeck, D.G., et al (2018). Characteristics, occurrence, and decay rates of remnant belts associated with three-belt events in the Earth's radiation belts. *Geophysical Research Letters*, 45, 12099-12107. doi:10.1029/2018GL080274
25. **Pinto, V.A.**, Kim, H.-J., Lyons, L.R., Bortnik, J. (2018). Interplanetary parameters leading to relativistic electron enhancement and persistent depletion events at geosynchronous orbit. *JGR: Space Physics*, 123, 1134-1145. doi:10.1002/2017JA024902
26. Antonova, E.E., Stepanova, M.V., Moya, P.S., **Pinto, V.A.**, et al (2018). Processes in auroral oval and outer electron radiation belt. *Earth, Planets and Space*, 70, 127. doi:10.1186/s40623-018-0898-1
27. Farías, C., **Pinto, V.A.**, Moya, P.S. (2017). What is the temperature of a moving body?. *Scientific Reports*, 7(1), 17657. doi:10.1038/s41598-017-17526-4
28. Moya, P.S., **Pinto, V.A.**, Sibeck, D.G., Kanekal, S.G., Baker, D.N. (2017). On the effect of geomagnetic storms on relativistic electrons in the outer radiation belt. *JGR: Space Physics*, 122, 11100-11108. doi:10.1002/2017JA024735
29. Kim, H.-J., Lyons, L., **Pinto, V.A.**, Wang, C.-P., Kim, K.-C. (2015). Revisit of relationship between geosynchronous relativistic electron enhancements and magnetic storms. *Geophysical Research Letters*, 42, 6155-6161. doi:10.1002/2015GL065192
30. Moya, P.S., **Pinto, V.A.**, Vinas, A.F., Sibeck, D.G., et al (2015). Weak kinetic Alfvén waves turbulence during the 14 November 2012 geomagnetic storm. *JGR: Space Physics*, 120, 5504-5523. doi:10.1002/2014JA020281

31. **Pinto, V.A.**, Stepanova, M., Antonova, E.E., Valdivia, J.A. (2011). Estimation of the eddy-diffusion coefficients in the plasma sheet using THEMIS satellite data. *Journal of Atmospheric and Solar-Terrestrial Physics*, 73(11-12), 1472-1477. doi:10.1016/j.jastp.2011.05.007
32. Stepanova, M., **Pinto, V.A.**, Valdivia, J.A., Antonova, E.E. (2011). Spatial distribution of the eddy diffusion coefficients in the plasma sheet during quiet time and substorms from THEMIS satellite data. *Journal of Geophysical Research*, 116, A00I24. doi:10.1029/2010JA015887

Proceedings, chapters and white papers

1. Antonova, E.E., ..., **Pinto, V.A.**, et al (2023). Formation of the Outer Radiation Belt: Adiabatic Effect and Stochastic Acceleration. *Problems of Geocosmos-2022, Springer Proceedings in Earth and Environmental Sciences*. doi:10.1007/978-3-031-40728-4_23
2. Turner, D.L., et al. (incl. **Pinto, V.A.**) (2023). Cross-scale physics in the radiation belts. *Bulletin of the AAS*, 55 (*Heliophysics Decadal Survey white paper*). doi:10.3847/25c2cfef.7637306e
3. Bashir, M.F., et al. (incl. **Pinto, V.A.**) (2023). Recognition for all: a way forward to enhance diversity, equity and inclusion in space physics. *Bulletin of the AAS*, 55 (*Heliophysics Decadal Survey white paper*). doi:10.3847/25c2cfef.bff53876
4. Reyes, P., **Pinto, V.A.**, Moya, P.S. (2019). Statistical analysis of geomagnetic storms and their relation with the solar cycle. *Proceedings of the International Astronomical Union*, 15(S354), 224-227. doi:10.1017/S1743921320000903
5. Silva, P., Farías, C., L'Huissier, P., **Pinto, V.**, Zambra, M., Soto, L. (2008). Hard X-ray measurement from a plasma focus of low energy. *Journal of Physics: Conference Series*, 134, 012045. doi:10.1088/1742-6596/134/1/012045

Invited talks

Aug 2026	GSST/CBERS-5 SolarAI Symposium on Machine Learning in Heliophysics and Space Weather. INPE, Brazil
Nov 2022	Predicting GICs using machine learning. Physics Department Colloquium, USACH · Santiago, Chile
Nov 2022	The Earth's radiation belts. International Space Science School (ISSS) · INPE, Brazil
Dec 2020	Recent results on the understanding of Earth's double outer radiation belt events and remnant belt confinement. AGU Fall Meeting · Virtual
Sep 2020	Forecasting geomagnetically induced currents at mid-latitudes using machine learning. UNH Postdoctoral Appreciation Week · Virtual

Conference presentations (selected)

Dec 2025	II Chilean Plasma Physics Congress. Chile
Aug 2025	III Workshop on Instrumentation in Geosciences. Chillán, Chile
Jan 2025	Analysis of solar wind properties associated with relativistic electron enhancement events at geostationary orbit. 17th Latin American Workshop on Plasma Physics · Santiago, Chile
Dec 2024	Evaluation of gap-filling for OMNI data: how much data is it safe to interpolate? (poster). AGU Annual Meeting · Washington, D.C., USA
Apr 2024	Forecasting ground magnetic field maximum perturbations in near-real time with deep learning models. XIV Conferencia Latinoamericana de Geofísica Espacial (COLAGE) · Monterrey, Mexico
Dec 2023	Forecasting 1-hour ahead ground magnetic field maximum perturbations with deep learning models (poster). AGU Fall Meeting · San Francisco, USA

Dec 2022	Forecasting ground magnetic perturbations at high and mid-latitudes using deep learning and near real-time solar wind data (poster). AGU Fall Meeting · Chicago, USA
Nov 2022	Forecasting ground magnetic perturbations using deep learning and near real-time data. XIII Conferencia Latinoamericana de Geofísica Espacial (COLAGE) · INPE, Brazil
Jan 2022	Evaluating near real-time ground magnetic field perturbations predictions using machine learning models. AMS Annual Meeting · Houston, USA

Teaching at USACH

Undergraduate	Electromagnetism (2023-2, 2024-2, 2025-2, 2026-2)
Undergraduate	Advanced Laboratory (2023-1, 2026-2)
Undergraduate	Introduction to Heliospheric Physics (2024-1, 2025-1, 2025-2, 2026-1)
Graduate	Introduction to Heliospheric Physics (2024-1, 2026-1)
Undergraduate	Astronomical Instrumentation (co-taught) (2025-1)
Undergraduate	Computational Physics II and III (2023-1, 2023-2)
Ph.D.	Topics I: Data processing and analysis for space sciences and geosciences (2022-2)

EARLIER TEACHING

2021	University of New Hampshire, Department of Physics , Adjunct faculty. PHYS 408 General Physics II (recitation)
2015-2019	UCLA, Department of Atmospheric and Oceanic Sciences , Teaching Assistant, Associate and Fellow. Climate change, air and water pollution, atmospheric science courses (AOS 1, 3, 101, 102, 104)
2010-2011	Universidad Católica del Maule , Instructor (half-time faculty). General physics, thermal physics, computational physics and integrated science courses
2010-2011	Universidad de Valparaíso, Escuela de Ingeniería (Santiago) , Part-time lecturer. General Physics II (lectures and lab), Physics I and III labs
2009	Liceo Experimental Manuel de Salas , Physics teacher

CURRICULUM DESIGN

- Redesign of the B.Sc. Astrophysics with Data Science curriculum (2023-2024).
- Course designs for Computational Physics I-IV, Electromagnetism, Introduction to Heliospheric Physics and Topics I.

Thesis and postdoctoral supervision

COMPLETED

2025	Yulissa Espitia , M.Sc. in Physics, USACH — Advisor
2025	Carlos Castillo , M.Sc. in Physics, USACH — Co-advisor
2024	Christopher Lara , M.Sc. in Physics, Universidad de Chile — Co-advisor
2025	Guillermo Caro , Physics Engineering, USACH — Advisor
2024	Yerko Jelcic , Physics Engineering, USACH — Advisor
2024	Javier Ruiz , Physics Engineering, USACH — Co-advisor

IN PROGRESS

Paula Reyes, M.Sc. in Physics, Universidad de Chile — Co-advisor
Dafne Muñoz, Astrophysics Engineering, USACH

Camila Peña, Physics Engineering, USACH

Dra. Bea Zenteno-Quinteros, FONDECYT Postdoctoral grant (ANID), 2025– — Sponsor

Dra. Matilde Coello, DICYT Postdoctoral grant (USACH), 2024–2026 — Sponsor

Thesis committees (examiner)

2026	Alejandro Zamorano, Ph.D., U. de Chile
2026	Eduardo Flández, Ph.D., U. de Chile
2025	Fernando Castillo Morales, M.Sc., USACH
2025	Mohammad Darehmoradi, M.Sc., USACH
2024	José M. Espinoza, M.Sc., USACH
2024	Tomás Zurita, M.Sc., U. de Chile
2024	Ángel Barra, Physics Engineering, USACH
2023	Javier Silva, M.Sc., U. de Chile
2023	Francisca Tapia, M.Sc., U. de Concepción
2022	José M. Espinoza, Physics Engineering, USACH

Service

EDITORIAL

2024–2027 **Associate Editor, Space Weather (AGU/Wiley)**. Appointed November 2024.

REVIEW PANELS

2021–2026 **NASA review panels**. Heliophysics Supporting Research (2021), HGIO (2023), Living With a Star (2024), Space Weather R2O2R (2025–2026).

REVIEWING

2019– **Journal referee**. Space Weather, JGR Space Physics, JGR Machine Learning and Computation, GRL, JSWSC, JASTP, Nature Communications, Radio Science, Annales Geophysicae, Universe, PLOS One, among others.

2021– **Proposal reviewer**. FONDECYT (ANID), DICYT (USACH), NSERC (Canada), INACH.

UNIVERSITY

2025– **Associate Chair for Teaching, Department of Physics, USACH**

2022– **Degree committee, B.Sc. Astrophysics with Data Science**. Including the curriculum redesign (2023–2024).

2024 **Acting Program Director, Astrophysics (July–August)**

2024–2025 **Coordinator of physics courses for the Faculty of Technology**

2024 **Teaching Committee, Department of Physics**

2022– **Faculty member, M.Sc. and Ph.D. programs, Faculty of Science**

2024–2025 **Reviewer, Faculty of Science internal fund**

SCIENTIFIC SOCIETIES

2021–2024 **AGU Space Physics and Aeronomy, Nominations Task Force**

2022–2024 **Chile representative, ALAGE (Latin American Association of Space Geophysics)**

2026– **Member, SOCHIAS (Chilean Astronomical Society)**

2015–2022 **Founding member, ANIP**

2020-2022	UNH Postdoctoral Association, webmaster and international affairs chair
2012-2017	UCLA AOS Graduate Students Association, president (2016-2017) and webmaster
2016-2017	UCLA Dashew Center, international ambassador (Chile, Argentina, Ecuador, Peru, Uruguay)

EVENTS ORGANIZED

Jun 2026	IGS Workshop 2026 (International GNSS Service). Organizing committee and session chair · Santiago, Chile
Aug 2025	III Workshop on Instrumentation in Geosciences. Co-organizer · Chillán, Chile
Sep 2023	Physics Graduate Research Days, USACH. Organizing committee · Santiago, Chile
May 2023	II Workshop on Instrumentation in Geosciences. Organizer · Santiago, Chile
Jan 2022	AMS Annual Meeting. Session proposer and chair · Houston, USA

Awards and honors

2023	Most read paper of 2021, Space Weather. “Geomagnetic storm occurrence and their relation with solar cycle phases”
2018	AGU Outstanding Student Paper Award (AGU Fall Meeting 2017). “Spatial characterization of relativistic electron events in the Earth's outer radiation belt”
2018	Top 100 Physics Papers 2017, Scientific Reports. “What is the temperature of a moving body?”
2011	Becas Chile doctoral fellowship (CONICYT)
2009	Thesis support in Antarctic topics (INACH)

Outreach and media (selected)

Besides press work, I have taught space weather workshops at the Faculty of Science Winter School (2023 and 2024), given talks at schools in Santiago and Maipú, and sponsored student outreach projects.

Apr 2026	Canal 13: Artemis II launch and lunar trajectory
Dec 2025	Diario Usach, Cooperativa Ciencia, Crónica Digital, Mundo en Línea, Tu Opinión: Launch of Chile's first space weather platform (FONDEF)
Nov 2025	Radio Bío-Bío (entrevista), BioBioChile, El Mercurio, Diario Usach, Diario Chiloé, Diario de Puerto Montt, Diario Palena, Página 7, Media Banco: Purple-toned southern auroras seen in southern Chile
Oct 2025	Mega: Weakening of Earth's magnetic field over the South Atlantic

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